MiniTec AI Strategy
Speaker Scripts & Q&A Prep

Presentation duration: **10–15 minutes** · Team size: **5 speakers**

Distribute this document to all team members before presenting.

TIMING OVERVIEW

Speaker 1	Speaker 2	Speaker 3	Speaker 4	Speaker 5
~2.5 min	~2.5 min	~3 min	~2.5 min	~3 min
Slides 1–2	Slides 3–4	Slides 5–7	Slides 8–9	Slides 10–12

SPEAKER 1

Introduction & Executive Summary

Slides 1–2 · ~2.5 min

SLIDE 1 — TITLE

Good afternoon, everyone. Thank you for having us. My name is [Name], and together with my team, we're here to present our AI strategy proposal for MiniTec.

The title of our presentation is: *"AI That Eliminates Wasted Engineering Hours"* — and that's really the core of what we're going to talk about today. We've identified two very specific bottlenecks in how we operate, and we're proposing three concrete AI solutions to fix them. I'll hand over to my team after this introduction, but let me start by framing the problem.

SLIDE 2 — EXECUTIVE SUMMARY

So what exactly are these two problems?

The first one we're calling the **Search Gap**. When an engineer needs a specific 3D model in our Autodesk Vault, they can only search by exact keywords. If the part was named differently, they don't find it — and so they redraw it from scratch. Studies show that up to 60% of new CAD parts created in manufacturing companies are actually duplicates of existing ones. That's over 1,200 hours per engineer per year, completely wasted.

The second problem is the **Documentation Bottleneck**. Right now, creating one operating manual requires someone to manually copy and paste data from four different systems — the CAD software, the ERP, the BOM, and old manuals. That takes between 40 and 80 hours per manual. And 80% of that content is basically the same across product variants.

Our proposal is to use AI to fix both of these. I'll now pass it over to [Speaker 2's name] to walk you through where the industry is right now and how we came up with our approach.

> **Tip:** Make eye contact when presenting the stats. Pause after "1,200 hours" to let it land before moving on.

SPEAKER 2

Status Quo & Ideation

Slides 3–4 · ~2.5 min

SLIDE 3 — STATUS QUO

Thanks [Speaker 1]. My name is [Name], and I'm going to give you a picture of where we are today versus where our competitors are.

Internally, our situation is quite manual. Vault is purely keyword-based — there's no intelligence behind the search. Our documentation process is completely copy-paste across systems. As I said, 80% of content is repetitive, but nobody has automated it yet.

Externally, though, the industry has already moved. Companies like CADENAS PARTsolutions have been offering AI-based shape and semantic search for CAD parts for several years now. Dassault's EXALEAD platform can automatically detect duplicate parts across entire CAD libraries. Cortona3D already generates 3D technical manuals directly from CAD models. And modern language model pipelines are cutting SOP writing time by 60 to 75 percent at companies that have adopted them.

The technology is not new or experimental — it's already in production at our competitors. We simply haven't adopted it yet.

SLIDE 4 — STRATEGY & IDEATION

So how did we decide what to do about it? We didn't just pick solutions off a list — we went through a structured ideation process.

We started by clearly defining the two problem areas. We then brainstormed over ten different AI ideas — everything from predictive maintenance to AI-driven procurement. We filtered those down by asking: which ones are directly relevant to our biggest pain points, and which ones are technically feasible right now?

After that filter, we applied a MoSCoW prioritization. The "Must Have" solution is a smarter search tool. The "Should Have" is automated manual generation. And the "Could Have" is a more advanced video-based documentation tool for the factory floor.

Our overarching strategy has two non-negotiable rules: every AI we deploy must *augment* our engineers — not replace them — and it must plug directly into systems we already use, like Vault and our ERP. No big rip-and-replace projects. I'll now pass it to [Speaker 3] to walk you through the first two use cases.

> **Tip:** On the pipeline diagram, point to each box as you describe the steps. It makes the ideation process feel structured and credible.

SPEAKER 3

Use Cases 1, 2 & 3

Slides 5–7 ~3 min

SLIDE 5 — USE CASE 1: TEXT-TO-CAD MATCHMAKER

Thanks. I'm [Name], and I'll walk you through the three AI solutions we're proposing.

The first is what we're calling the **Text-to-CAD Matchmaker**. The idea is simple: an engineer types what they need in plain English — something like "stainless steel bracket, 500 kilograms, three bolt holes" — and the AI understands that request and scans our entire Vault library to find the closest matching part. Right now, that would only work if someone happened to name the file with those exact words. With this AI, the system understands the meaning, not just the words.

There's also a built-in duplicate detection feature. Before an engineer saves a new part, the AI automatically checks whether something similar already exists and warns them. This alone could stop the majority of unnecessary duplicates from being created.

SLIDE 6 — USE CASE 2: AUTO MANUAL GENERATOR

The second use case tackles the documentation bottleneck directly. We call it the **Automatic Manual Generator**.

Instead of a technical writer spending 40 to 80 hours collecting information and drafting a manual from scratch, this system does the heavy lifting automatically. It connects to Vault, the BOM, and our ERP, pulls the relevant data, and uses a language model to write the full first draft of the manual. The writer then reviews, adjusts, and approves. Their job shifts from drafter to editor — which is a much more valuable use of their time.

And I want to be clear — this is not about removing human judgment. Any safety-critical content in the manual must still be reviewed and signed off by an engineer. The AI is the drafter. The human is always the final approver.

SLIDE 7 — USE CASE 3: WATCH & WRITE VIDEO AI

The third use case is the most innovative. It's called **"Watch and Write"**, and it uses computer vision to generate assembly instructions automatically.

Here's how it works: a technician on the factory floor wears smart glasses or simply uses a phone to record themselves performing an assembly. While they work, they narrate what they're doing. The AI watches the video, identifies each action step using computer vision, transcribes the speech, and automatically produces a written step-by-step guide — complete with screenshots from the recording.

The entire instruction manual gets created without anyone sitting down to type a single word. This is

especially powerful for complex or legacy processes that have never been documented before. I'll now hand it to [Speaker 4] for our evaluation.

> **Tip:** Keep UC1 and UC2 tight — about 45 seconds each. Give UC3 a bit more time since it's the most visual and innovative concept.

SPEAKER 4 Prioritization & Financials

Slides 8–9 ~2.5 min

SLIDE 8 — PRIORITIZATION MATRIX

Thanks [Speaker 3]. My name is [Name]. Now that you've seen what the three use cases do, I want to show you how we evaluated and ranked them.

We scored each use case across four criteria: Strategic Impact, Implementation Feasibility, Time to Value, and AI Innovation — each weighted based on what matters most for a company like MiniTec right now.

The Text-to-CAD Matchmaker scores highest overall at 4.35 out of 5. It has maximum strategic impact, it's the most feasible to implement using existing Vault APIs, and it delivers value the fastest. That's why it's our first priority and where we want to start.

The Auto Manual Generator scores 3.9 — also excellent, and it's sequenced second. The Video AI scores 3.05, which is still strong, but it requires more hardware investment and time to deploy, so it's our Phase 3 target.

SLIDE 9 — FINANCIAL EVALUATION

And here's the financial case for each one.

The Matchmaker requires roughly €30,000 in initial development plus around €12,000 per year in cloud compute. With over 1,100 hours of engineering time saved annually and a 90% reduction in duplicate part creation, it pays for itself in about twelve months.

The Auto Manual Generator costs slightly more upfront — around €45,000 — but by cutting manual creation time from 80 hours down to 2, it pays back in roughly 15 months.

The Video AI is the biggest investment, approximately €80,000 for hardware and development, with a 24-month payback. But the long-term return is significant — near-zero documentation time for assembly processes and a major reduction in instruction errors.

All three use cases pay for themselves well within two years. This isn't just an innovation project — it's a sound financial investment. I'll hand it to [Speaker 5] to close out with risks, roadmap, and next steps.

> **Tip:** On the financial slide, briefly point to the bar charts to make them feel concrete. Don't just read numbers — connect them to the real work impact.

SPEAKER 5

Risks, Roadmap & Conclusion

Slides 10–12 · ~3 min

SLIDE 10 — RISK MANAGEMENT

Thank you. My name is [Name], and I want to make sure we're transparent about the risks — because any honest proposal has to address them.

The two highest-risk areas are legal and safety. On the legal side, we cannot send our proprietary CAD designs to a public AI service like ChatGPT or open AI APIs. All AI processing must happen on-premise or in a private enterprise cloud where no data leaves our organization. That's a hard requirement.

On the safety side — particularly for Use Case 2 — there's a risk that an AI generates an incorrect safety instruction in a manual. To prevent that, we've built in a mandatory human review step before any AI-generated manual can be published. The engineer is always the final approver.

We also acknowledge the ethical dimension. Some team members may worry that AI means their job is at risk. Our clear message is that it doesn't — AI handles the repetitive data-gathering. People handle judgment, quality control, and complex decisions. That's a much better use of skilled engineers.

SLIDE 11 — ROADMAP

Here's how we'd actually do this, in three realistic phases.

In **Phase 1**, months one to six, we start by cleaning up our Vault metadata — because AI search is only as good as the data underneath it. Then we deploy the Text-to-CAD Matchmaker and run it in parallel with the existing system. We expect visible ROI within the first quarter after go-live.

In **Phase 2**, months seven to twelve, we create the standard documentation templates and deploy the Auto Manual Generator. We pilot it on a small set of products first and scale based on feedback.

In **Phase 3**, months thirteen to eighteen, we pilot the Watch and Write video system on the factory floor — one product line, one team — and evaluate before expanding further.

SLIDE 12 — CONCLUSION

To wrap up — MiniTec's motto is "The Art of Simplicity." That's exactly what this strategy is about. Let AI handle the repetitive, so our people can handle the complex.

The results we're targeting are concrete: a 90% reduction in duplicate parts, cutting manual creation from 80 hours down to 2, increasing our content reuse by 800%, and seeing the first financial payback in around twelve months.

We're not proposing a science experiment. This technology is proven, our competitors are already using it, and the financial case is solid. We're asking for approval to move into Phase 1 and kick off a four-week proof of concept with the Matchmaker.

Thank you all for your time. We're happy to take any questions.

> **Tip:** End on the quote — pause before you say it and deliver it with confidence. It's a strong close. Then smile and say "We're happy to take any questions."

💬 Q&A Preparation — Likely Questions & Answers

Q: How do you know 60% of parts are duplicates? That seems very high.

That statistic comes from the Aberdeen Group's 2022 manufacturing study. It's an industry average, not a MiniTec-specific figure — so our actual number could be lower. But even if it's 30%, that still represents hundreds of wasted engineering hours per year. The point is the scale of the problem, not the exact number.

Q: What if the AI gives wrong results in the search? Engineers might trust it too much.

That's a really valid point. The system always shows a confidence score with each result, and it's designed as a suggestion tool — not a replacement for the engineer's judgment. Engineers still verify before using a part. The AI saves them time finding candidates. The final decision is always theirs.

Q: Are you worried about AI writing wrong safety instructions in manuals?

Absolutely — and that's exactly why we built in a mandatory human review step. No AI-generated manual gets published without an engineer signing off on it, especially safety-critical content. The AI is the first draft writer, the human is the editor and approver. This is also how CE and ISO compliance is maintained.

Q: What happens to the technical writers? Will they lose their jobs?

No — their role changes, it doesn't disappear. Instead of spending 80% of their time copy-pasting data, they focus on reviewing, improving quality, handling edge cases, and managing the AI's output. That's actually a more skilled and more satisfying job. We're also proposing an upskilling program as part of implementation.

Q: Is this secure? We have confidential CAD designs.

Security is a non-negotiable requirement for us. All AI processing will run on-premise or in a private enterprise cloud. No design data goes to any public AI service. We are specifically ruling out any solution that requires sending data externally.

Q: Have you tested any of this? Is it proven technology?

The technology itself is very much proven — CADENAS, EXALEAD, and Cortona3D have been in production at large manufacturing companies for years. What we're proposing in Phase 1 is a four-

week proof of concept specifically for MiniTec's Vault setup, so we validate the integration before committing to a full rollout. Low risk, high learning.

Q: Why not just buy an off-the-shelf solution instead of building one?

That's actually still on the table, particularly for Use Case 1. CADENAS PARTsolutions could be licensed as a COTS solution in 3 to 4 months rather than building from scratch. Part of the Phase 1 proof of concept is to evaluate exactly that — build versus buy — and recommend the most cost-effective path.

Q: €80,000 for the Video AI is a lot. Is that realistic?

The €80,000 estimate includes hardware — smart glasses or camera rigs — plus custom computer vision development. That's why it's our Phase 3 item, after we've already seen ROI from Phases 1 and 2. By the time we get there, we'll have strong evidence that AI investment pays off at MiniTec, which makes the business case much easier to approve.